# Procurement Restructuring & Supplier Intelligence Dossier

Tender UP-10A6DB37 - Protection Work of Damaged Portion near RT Wall of Diversion Channel of Ukai Dam.
Generated 2026-11-13T04:41:31+00:00 by TenderShield Nexus (decision support - not an award decision)

## Consider 3 packages instead of one contract.

## A. Original procurement

Estimated value: Rs 46.84 Cr. Current structure: 1 package(s): Civil & building construction + Electrical systems & power + Operation & maintenance services + Telecom & communication networks.

| ID | Component | Items | Value | Value basis | Source pages |
|---|---|---|---|---|---|
| C1 | Civil & building construction | 4 | Rs 25.06 Cr | document | 1 |
| C2 | Electrical systems & power | 1 | Rs 6.50 Cr | document | 1 |
| C3 | Telecom & communication networks | 1 | Rs 2.38 Cr | document | 1 |
| C4 | Operation & maintenance services | 1 | Rs 12.89 Cr | document | 1 |

## B. Proposed restructuring

3 package(s): Civil & building construction | Electrical systems & power | Operation & maintenance services + Telecom & communication networks

Interfaces the employer must manage if separated:

| Interface | Weight | Type |
|---|---|---|
| C1 - C2 | 0.35 | installation |
| C1 - C3 | 0.3 | installation |
| C1 - C4 | 0.3 | schedule |
| C2 - C3 | 0.15 | schedule |
| C2 - C4 | 0.3 | schedule |

## C. SME opportunity map

| Pkg | Scope | Value | Turnover req. | Capable firms | Meet scale rule | SME candidates | Bidders (ind.) |
|---|---|---|---|---|---|---|---|
| P1 | Civil & building construction | Rs 25.06 Cr | Rs 7.52 Cr | 1119 | 373 | 795 | 2.792 |
| P2 | Electrical systems & power | Rs 6.50 Cr | Rs 1.95 Cr | 230 | 141 | 162 | 2.361 |
| P3 | Operation & maintenance services + Telecom & communication networks | Rs 15.27 Cr | Rs 4.58 Cr | 6 | 5 | 2 | 0.225 |

P1 potential supplier pool (top 6 of 1119)

| Supplier | Country | Status | MSME status | Score | Evidence |
|---|---|---|---|---|---|
| A. K. PATEL AND COMPANY [CONSTRUCTION] PVT. LTD., SURAT | India | POTENTIALLY_SUITABLE | SIZE_PROXY_SMALL | 0.9 | 6 awards in Civil & building construction (largest [18.449]) |
| AMAR CONSTRUCTION CO. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 5 awards in Civil & building construction (largest [33.149]) |
| BACKBONE ENTERPRISES LTD. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 7 awards in Civil & building construction (largest [371.604]) |
| M/S CHANDRAGIRI CONSTRUCTION | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 8 awards in Civil & building construction (largest [37.128]) |
| FCI ARAVALI GYPSUM MINERALS INDIA LTD. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 7 awards in Civil & building construction (largest [101.552]) |
| M/S GANGOTRI ENTERPRISES LTD., LKO | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 8 awards in Civil & building construction (largest [75.583]) |

P2 potential supplier pool (top 6 of 230)

| Supplier | Country | Status | MSME status | Score | Evidence |
|---|---|---|---|---|---|
| LARSEN AND TOUBRO LIMITED | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 1.0 | 9 awards in Electrical systems & power (largest [251.015]) |
| ABB LIMITED | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 18 awards in Electrical systems & power (largest [198.894]) |
| ALSTOM LTD. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 9 awards in Electrical systems & power (largest [49.59]) |
| M/S ALSTOM LTD., JAIPUR | India | POTENTIALLY_SUITABLE | SIZE_PROXY_SMALL | 0.9 | 8 awards in Electrical systems & power (largest [13.006]) |
| ALSTOM T&D INDIA LIMITED | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.9 | 7 awards in Electrical systems & power (largest [396.243]) |
| BHEL | India | POTENTIALLY_SUITABLE | SIZE_PROXY_SMALL | 0.9 | 5 awards in Electrical systems & power (largest [23.791]) |

P3 potential supplier pool (top 6 of 259)

| Supplier | Country | Status | MSME status | Score | Evidence |
|---|---|---|---|---|---|
| LARSEN AND TOUBRO LIMITED | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.808 | 1 awards in Telecom & communication networks (largest [360.481]); 3 awards in Operation & maintenance services (largest [344.417]) |
| CMC | India | POTENTIALLY_SUITABLE | SIZE_PROXY_SMALL | 0.74 | 1 awards in Telecom & communication networks (largest [3.173]); 1 awards in Operation & maintenance services (largest [14.255]) |
| N.G. PROJECTS LTD. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.74 | 1 awards in Telecom & communication networks (largest [85.663]); 1 awards in Operation & maintenance services (largest [29.783]) |
| P AND C CONSTRUCTIONS LTD. | India | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.74 | 1 awards in Telecom & communication networks (largest [3.514]); 1 awards in Operation & maintenance services (largest [21.084]) |
| ASSIGNIA INFRAESTRUCUTURAS | Spain | POTENTIALLY_SUITABLE | SIZE_PROXY_LARGE | 0.74 | 1 awards in Telecom & communication networks (largest [86.296]); 1 awards in Operation & maintenance services (largest [227.562]) |
| S A YADAV | India | PARTIAL_CAPABILITY | SIZE_PROXY_SMALL | 0.726 | 4 awards in Operation & maintenance services (largest [10.249]) |

## D. Supplier relationships (secondary objective)

28 shortlisted suppliers screened; 0 relationship(s) require review. Signals for human review only. Relationships are not evidence of wrongdoing.

| Supplier A | Supplier B | Observed link | Review score | Ordinary explanations |
|---|---|---|---|---|
| AMAR CONSTRUCTION CO. | AMR CONSTRUCTIONS LIMITED | name_overlap | 0.2 | Similar names can be unrelated firms; confirm with registry identifiers (CIN/GSTIN) |
| A. K. PATEL AND COMPANY [CONSTRUCTION] PVT. LTD., SURAT | LARSEN AND TOUBRO LIMITED | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| A. K. PATEL AND COMPANY [CONSTRUCTION] PVT. LTD., SURAT | AMAR CONSTRUCTION CO. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |

| Supplier A | Supplier B | Observed link | Review score | Ordinary explanations |
|---|---|---|---|---|
| ABB LIMITED | ALSTOM LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ABB LIMITED | BHEL | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ABB LIMITED | CROMPTON GREAVES LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ABB LIMITED | HINDUSTAN VIDYUT PRODUCTS LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ALSTOM LTD. | BHEL | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ALSTOM LTD. | CROMPTON GREAVES LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| ALSTOM LTD. | HINDUSTAN VIDYUT PRODUCTS LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| BHEL | CROMPTON GREAVES LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |
| BHEL | HINDUSTAN VIDYUT PRODUCTS LTD. | same_project | 0.1 | Large projects legitimately award separate packages to different firms |

## E. Scenario comparison

| Scenario | Packages | Score | Competition | Coverage | SME access | Coordination | Integration risk | Scale loss |
|---|---|---|---|---|---|---|---|---|
| A | 1 | 0.0934 | 0.0502 | 0.3333 | 0 | 0.0 | 0.0 | 0 |
| B | 3 | 0.4893 | 0.3791 | 1.0 | 1.0 | 0.5667 | 0.1989 | 0 |
| C | 4 | 0.4734 | 0.5118 | 1.0 | 1.0 | 0.7667 | 0.3489 | 0 |

Weight sensitivity: the recommended structure stays best in 89% of 300 runs with every weight perturbed (each weight x U(0.5, 1.5)). Weights: {"competition": 0.2, "coordination": 0.1, "coverage": 0.25, "integration_risk": 0.15, "scale_loss": 0.05, "sme_access": 0.25}

## F. Recommendation

Decision support only. The procurement authority decides the structure, verifies supplier evidence and MSME status, and reviews any flagged relationships.

[INFO] Low-confidence capability classification - Check the listed source items

[INFO] MSME status shown as a size proxy unless Udyam evidence was imported - Import verified Udyam records via POST /api/market/udyam

## Data sources and models

Market scope: ['India']; 7434 award records in scope from worldbank_awards:major_contract_awards:india, prozorro:ua.

capability_classifier: TF-IDF (word 1-2g + char 3-5g) + multinomial logistic regression, lexicon-blended at inference. Trained 2026-11-13T04:35:49+00:00 on Common Procurement Vocabulary (CPV 2008) (9454 rows); Annotated tender lines (53 rows). Metrics: {"cv_accuracy": 0.7875, "cv_folds": 5, "cv_macro_f1": 0.7383}

- CPV descriptions are short catalogue phrases; long Indian BOQ lines are a domain shift - the lexicon blend and the annotated-line evaluation exist for this reason.

- Single label per text; a mixed BOQ line is assigned its dominant capability.

competition_model: HistGradientBoostingRegressor (Poisson loss) on log10(value Rs Cr) + capability one-hot. Trained 2026-11-13T04:35:51+00:00 on Prozorro lots (Ukraine public procurement, 2019-2023) (92438 rows). Metrics: {"mae": 1.0558, "mae_baseline": 1.064, "mean_bidders_test": 1.979, "poisson_deviance": 0.6907, "poisson_deviance_baseline": 0.7008, "split": "time: train 2019-2021, test 2022+", "test_rows": 14170, "train_rows": 78268}

- Learned on Ukrainian open tenders; used as a relative indicator between scenarios, not a forecast of Indian bidder counts. Recalibrate with CPPP bid-opening data.

- Lots in the data carry one CPV division, so the model has no notion of bundling; the bundling penalty comes from real supplier co-capability evidence instead.

- Observational data: larger lots in this data attract slightly more bidders; that is a correlation (buyers, procedures and markets differ), not a causal effect of lot size.

Capability taxonomy keys: civil_works = Civil & building construction, electrical = Electrical systems & power, signalling_control = Signalling, traffic & safety control, instrumentation_sensors = Instrumentation, sensors & metering, ict_software = Software, ICT & control systems, telecom_networks = Telecom & communication networks, mechanical_hvac_plumbing = Mechanical, HVAC, pumps & plumbing, industrial_machinery = Industrial & process machinery, structural_fabrication = Structural steel & specialised fabrication, consultancy_design = Design, engineering & supervision services, operation_maintenance = Operation & maintenance services, vehicles_equipment = Vehicles & heavy equipment, medical_lab = Medical & laboratory equipment, other_goods_services = Other goods & services